

ANAYA JAIN

Aspiring AI and Computer Science Researcher

<https://anayajain.me/>

Pune, India

SUMMARY

Grade 12 student with a strong foundation in Computer Science, AI, and Mathematics. Committed to expanding access to STEM education through teaching, mentorship, and community-building. Seeking to pursue Computer Science and AI at the undergraduate level, with an emphasis on building technology and doing research that creates meaningful impact.

EDUCATION

Grade 12 (CBSE)

Apr '27

EON Gyanankur English School, Pune, India

Grade 9, 10 (CISCE)

Apr '25

Vibgyor High, Magarpatta, Pune, India

Final Grade: 97%

SKILLS

Programming Languages & Technologies: Python, C, Java, JavaScript, SQL, HTML, CSS, TensorFlow, Deep Learning, CNNs, DenseNet-121, Image Classification, Grad-CAM, Data Analysis, Flask, Responsive Web Design, ThingSpeak, ESP32, GitHub, Sensor Integration, Data Structures, Algorithms, Object-Oriented Programming, Database Management, Mathematical Problem-Solving

Coursework & Certifications: Harvard CS50x: Introduction to Computer Science; AP Computer Science A; AP Calculus AB

Analytical Skills: Algorithms, Data Structures, Mathematical Problem Solving, Data Analysis

ACADEMIC and RESEARCH PROJECTS

Rice Leaf Disease Detection using Deep Learning

June '25 - Present

Research Poster Presenter, IEEE MIT URTC 2025

- Analyzed lesion severity across 10,407 rice leaf images using a DenseNet-121 model paired with Grad-CAM heatmaps to identify and quantify diseased regions.
- Presented the work as a poster at the IEEE MIT Undergraduate Research Technology Conference (URTC) 2025.

Research Author, Journal of Emerging Investigators

- Proposed integrating the approach into drone-based precision agriculture systems to support scalable, early-intervention disease monitoring for farms.
- Paper accepted in Journal of Emerging Investigators (JEI), passed scientific review, and is under copyediting

Air Atlas: Hyperlocal Air Quality Mapping

Mar '26 – May '26

IoT-based environmental sensing project developed under CERN mentorship

- Built and deployed an ESP32-based IoT air-quality monitoring node in Pune, measuring PM1, PM2.5, PM10, temperature, humidity, and atmospheric pressure.
- Monitored sensor readings through ThingSpeak for one month, collecting continuous data from the Pune node.
- Contributed Pune node data to Air Atlas, a live dashboard that aggregates measurements from 10 locations across India and visualizes AQI trends, pollution spikes, temperature, humidity, and other environmental patterns.
- Collaborated with team members operating nodes in different locations to create a hyperlocal, multi-location view of air quality across India.

Guitar Chord Shape Identifier

Mar '26

Kode with Klossy CODE-A-BRATION project

- Built a website and developed an image classification model to identify guitar chord shapes, helping beginner guitarists visualize correct finger placement.

Learning Buddy

June '25 - Oct '25

Harvard's CS50x final project

- Developed a web platform connecting underserved students with volunteer tutors using HTML, CSS, JavaScript, Python (Flask), and SQLite.
- Applied CS50x concepts, including data structures, algorithms, database management, and backend development, to implement user registration, tutor matching, and session scheduling.

EXPERIENCE

New York Academy of Sciences – Junior Academy Member

Sep '26 – Present

- Selected for the Junior Academy, a global community connecting high school students with STEM opportunities, mentorship, and research experiences.
- Collaborating on a six-person team for NeuroPulse, a wearable neurotechnology project exploring how ear-EEG, PPG, and HRV signals integrate to reveal patterns associated with atrial fibrillation (AFib).
- Exploring the software side of the project, focusing on analyzing multimodal physiological signals and their potential applications in long-term, non-invasive AFib monitoring.

Kode With Klossy Summer Camp - Web Development

Aug 2026

- Completed an intensive web development program focused on HTML, CSS, JavaScript, and responsive web design.
- Built hands-on web projects, "Guitar Chord Shape Identifier" while collaborating with peers in a women-in-tech learning community.

Girls Who Code Summer Pathways - AI & Data Science; AI+Society

Jun '26 – Aug '26

- Engaged in comprehensive studies of data analytics and intelligent systems through practical software projects
- Investigated the intersection of technology and ethics, focusing on the global effects of artificial intelligence.

Leaf Independent Learner

Apr '26 - May '26

- Selected as an Independent Learner in Leaf's five-week program, independently studying The Mathematics of Morality and Dilemmas and Dangers in AI through self-directed readings and coursework.
- Explored moral decision-making, mathematical approaches to ethics, AI alignment and safety, AI risks, and the societal implications of increasingly capable AI systems, developing independent perspectives on complex ethical questions.

HERizon: Founder and Lead

Mar '26 – Present

<https://herizonadmin.wixsite.com/herizon>

- Founded a student-led initiative to empower girls in STEM, reaching 100+ students to date.
- Organized monthly webinars with industry leaders and collaborated with organizations across India and Kenya to expand outreach.

Schoolhouse.world: Volunteer Tutor Math & President, Dialogues Club

Jan '26 – Present

[Free Online Tutoring — Schoolhouse](#)

- Hosted 22 SAT Math, Pre-Algebra, and Calculus sessions, impacting 113 learners across 25 countries and logging 1,200+ tutoring minutes.
- Conducted live SAT Math Bootcamps and tutoring sessions in Pre-Algebra and Calculus.
- Served as Summer and Fall Dialogues Club President.

Harvard CS50x: Introduction to Computer Science*June '25 – Oct '25*

- Introduction to Computer Science, building a foundation in algorithms, data structures, and problem-solving through nine programming problem sets in C, Python, and SQL.
- Built a web platform connecting underprivileged students with volunteer tutors, using HTML, CSS, JavaScript, and Python (Flask) with a SQL database to handle user sign-ups, tutor matching, and session scheduling.

Lodha Genius Programme Scholar, Ashoka University*May '25 – Dec '25*

- Selected among 8,000 applicants nationwide, placing in the top 4% of students selected in the country.
- Completed advanced coursework in Pure Mathematics, including Number Theory, Linear Algebra, Geometry, and Group Theory.
- Completed the online continued-learning module in the five-month Physics of Stars course; selected as one of the top 5 students in my course to visit NCRA-TIFR and the Giant Metrewave Radio Telescope (GMRT).
- Selected as a participant in the CERN International Masterclass, analyzing real particle physics data and studying experiments such as ALICE, CMS, and Antimatter Factory; attended Nobel laureate seminars; led 20 students as Ambassador.

Vibgyor High: Student Council Editor*Jun '23 – Mar '24*

- Led content planning, editing, and publishing for the school's official magazine publication.
- Developed strong communication and leadership skills through team coordination and editorial decision-making.

Model United Nations – Intermediate Level*Jun '23 – Mar '24*

- Represented China in the UN General Assembly at VIBGYOR MUN, debating the Russia-Ukraine war and researching China's position on the conflict; earned a Special Mention for diplomacy, research, and public speaking.

Vibgyor High: Social Service Responsibility Cell*Jun'23–Mar '24*

- Organized and led community outreach initiatives focused on education, raising over ₹7000 to support underprivileged students.
- Participated in donation drives and awareness campaigns while mentoring junior members and promoting social responsibility among peers.

INTERESTS & EXTRACURRICULAR ACTIVITIES

John Locke Institute 2026 Global Essay Prize - Commendation*April - Oct 2026*

- Awarded Commendation in Science & Technology among 100,227 entries for "Is free speech the enemy of science?"; invited to the 2026 Awards Dinner in London.

Trinity College London – Acoustic Guitar*Mar '21 – Present*

- Completed Initial, Grade 1, and Grade 3 with distinction; currently preparing for Grade 5, developing strong musical skills and discipline through structured training and examinations.

LifeTide Blog: <https://lifetide01.wordpress.com/>*2025 - Present*

- Publish reflective essays exploring science, curiosity, and personal growth.

School Dance Team*2023 – 2024*

- Led the senior secondary dance team.
- Participated in and choreographed school performances and annual productions.